MILITARY SPECIFICATION

HOSE ASSEMBLY, METAL, TRANSFER, LIQUID OXYGEN

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 This specification covers one type of liquid oxygen, transfer, metal hose assembly.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications form a part of this specification. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply.

SPECIFICATIONS

Federal	
BB-N-411	Nitrogen
O-P-191	Perchloroethylene (Tetrachlorethylene). Technical Grade
O-T-634	Trichloroethylene, Technical
Military	
MIL-P-l16	Preservation, Methods of
MIL-B-5087	Bonding, Electrical (For Aircraft)
MIL-T-5542	Thread Compound, Antiseize and Sealing, Oxygen Systems
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery

FSC 1660

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Military (Continued)

MIL-O-21749	Oxygen, Aviator's Breathing
MIL-F-22191	Films, Transparent. Flexible, Heat Sealing for Packaging Applications
MIL-B-22205	Bags, Transparent Flexible, Heat Sealing, for Packaging Applications
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MS33586	Metals, Definition of Dissimilar

Air Force-Navy Aeronautical Design

AND 10375	Colors- Fluid Line Identification

DRAWINGS

Air Crew Equipment Laboratory

33H1261	Hose Assembly, Metal, Transfer, Liquid Oxygen
33C1262	Adapter, Hose Assembly, Metal, Transfer, Liquid Oxygen

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government quality control representative. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding

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Officer, Naval Supply Depot (Code DCI), 5801 Tabor Avenue, Philadelphia 20, Pennsylvania, All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification - The hose assembly furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Selection of specification and standards - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with Standard MIL-STD-143.

3.3 Materials - Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings. Materials which are not covered by specifications. or which are not specifically described herein. shall be of the best quality. of the lightest practicable weight, and suitable for the purpose intended.

3.3.1 Metal parts - All metal parts shall be of a corrosion-resistant material or treated in a manner to render them adequately resistant to corrosion.

3.3.1.1 Dissimilar metals - Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in Standard MS33586.

3.3.2 Nonmetallic materials - Any nonmetallic material that is adversely affected by continued use with liquid oxygen shall not be used.

3.3.3 Protective treatment - When materials are used in the construction of the hose assembly that fire subject to deterioration when exposed to environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coating which might crack, chip, or scale during normal service life or under extremes of environmental conditions shall not be used.

3.4 Design and construction - The design and construction of the hose assembly shall be in accordance with ACEL Drawings 33H1261 and 33C1262.

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3.4.1 Antisieze compound - Antiseize and sealing compounds shall be used on all male pipe thread fittings by application of the compound to the third, fourth, and fifth threads (first and second threads shall be clean) from the end of the fittings. Compound shall not be used on flare tube fitting straight threads, coupling sleeves, or on the outside of tube flares. None of the compound shall be allowed to enter the inside of a fitting. Antiseize and sealing compounds shall conform to Specification MIL-T-5542.

3.5 Cleanliness - To remove oil, grease. residues, etc. from the applicable metal parts of the hose. externally and internally, the following procedure shall be followed. All applicable metal parts of the hose shall be:

a. degreased using a vapor phase degreaser in accordance with Specification O-P-191 or Specification 0-T-634.

b. flushed using trichloroethylene for a minimum of one minute.

c. oven dried at 250°F for a minimum of one hour.

d. immersed in Kelite 235 solution or equivalent for a minimum of five minutes.

e. rinsed with tap water for a minimum of five minutes.

f. oven dried at 250°F for a minimum of one hour.

g. examined for volatile residue presence by use of General Electric Type H halide detector or its equivalent. There shall not be any evidence of halides present,

h, examined for particle residue in the following reamer:

(1) The hose shall be 75% filled with distilled water at 160° F and both ends shall be sealed. The hose shall be shaken for a minimum of two minutes and the contents shall then be poured into a 100 milliliter dean glass cylinder. The contents shall not contain precipitates or be more turbid than a standard suspension of 5 milligrams of fullers earth in 100 milliliters of distilled water.

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i. purged and dried by passing hot (160°F maximum) dry oxygen conforming to Specification MIL-O-21749, type I, or nitrogen conforming to Specification BB-N-411, type I, class I, grade B, through the assembly.

j. capped with suitable end caps.

3.6 Performance - The hose assembly shall pass the performance requirements specified in Section 4 when subjected to the applicable tests.

3.7 Interchangeability - All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.8 Identification of product - The hose assembly shall be marked for identification in accordance with Standard MIL-STD-130. The hose shall also be marked in accordance with Standard AND 10375 and banded in accordance with Specification MIL-B-5087.

3.9 Workmanship - The hose assemblies shall be uniform in quality and shall be free from irregularities or defects which could adversely affect performance, reliability, or durability. The assemblies shall be free of oil, grease, fuel, water, dust, dirt, or any other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government, The Government rewerves the right to perform any of the inspections set forth in the specification where such inspections are deemed neces - sary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection - The examination and testing of the hose assemblies shall be classified as follows:

(a) Qualification inspection

(b) Preproduction inspection

(c) Quality conformance inspection

4.3 Qualification inspection - Qualification inspection shall consist of all the examinations and tests of this specification.

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4.3.1 Qualification samples - Qualification samples shall be accompanied by the manufacturer's inspection reports. Qualification samples shall consist of three complete hose assemblies. Samples shall be forwarded to the Supply Officer, Naval Air Engineering Center, Philadelphia 12, Pennsylvania, Attention: Director, Air Crew Equipment Laboratory. The samples shall be plainly identified by securely attached durable tags marked with the following information:

Sample submitted by (name) (date) for qualification inspection in accordance with the requirements of Specification MIL-H- (Wep), under authorization (reference Bureau of Naval Weapons letter authorizing the inspection).

4.3.1.1 Manufacturer's inspection report - Prior to forwarding qualification samples to the inspecting activity, additional samples shall have been inspected by the manufacturer, or an approved inspecting facility, in accordance with this specification. The report shall include the results of all the examinations and tests and a detailed statement indicating compliance or extent of non-compliance with each requirement of this specification, referring specifically to the applicable paragraph numbers in this specification.

4.4 Preproduction inspection - Preproduction inspection shall consist of all the examinations and tests of this specification.

4.4.1 Preproduction samples - Unless otherwise specified, as soon as practicable after award of a contract or order, the contractor shall furnish preproduction samples for inspection to determine conformance with this specification. The number of samples and the forwarding of the samples shall be as specified in 4.3.1 except that the contract number and manufacturer's part number shall be included. Preproduction inspection may include any or all of the inspections of this specification. Quantity production shall be withheld until the preproduction sample has been pronounced satisfactory by the Government. Preproduction inspection may be waived at the discretion of the Government. The approval of the preproduction samples or the waiving of the preproduction inspection shall not relieve the contractor of his obligation to submit samples for quality conformance inspection,

4.4.2 Upon completion of the preproduction inspection, all the applicable inspection reports, recommendations, and comments pertinent for use in monitoring production shall be forwarded to the Government quality control representative. Two hose assemblies shall be returned to the manufacturer for use in monitoring production. The other hose assembly shall be retained by the Air Crew Equipment Laboratory for reference.

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4.5 Quality conformance inspection - The quality conformance inspection shall consist of the following:

Visual examination (See classification of defects)
Dimensional check
Proof pressure
Obstruction
Endurance
impulse
Preparation for delivery

4.5.1 Lot -

4.5.1.1 Hose assembly - An inspection lot size shall be expressed in units of hose assemblies, made essentially under the same conditions and from the same materials and components. The sample unit shall be one hose assembly.

4.5.1.2 Preparation for delivery - An inspection lot size shall be expressed in units of one fully prepared shipping container, containing hose assemblies made essentially from the same materials and components, The sample unit shall be one shipping container, fully prepared for delivery with the exception that it need not be sealed.

4.5.2 Sampling -

4.5.2.1 Sampling for tests and examinations of hose assemblies - The sample size, acceptance criteria, tests, and examinations required for the hose assemblies shall be as specified in Table I and 4.5.2.2.

TABLE I

SAMPLE SIZE, ACCEPTANCE CRITERIA. TESTS, AND EXAMINAT1ONS OF THE HOSE ASSEMBLIES

INSPECTION	TYPE OF INSPECTION	METHOD	SAMPLE SIZE	ACCEPTANCE CRITERIA
Visual examination	Major	4.7.1	Every hose assembly for critical defects. Inspection Level II 1/ for minor defects.	Acceptance number zero, rejection number 1 for critical defects and an accepted quality level of 10.0 defects per hundred units for minor defects.

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TABLE I (Continued)

SAMPLE SIZE, ACCEPTANCE CRITERIA, TESTS, AND EXAMINATIONS OF THE HOSE ASSEMBLIES

INSPECTION	TYPE OF INSPECTION	METHOD	SAMPLE SIZE	ACCEPTANCE CRITERIA
Dimensional check	Major	4.7.1.1	Every hose	Acceptance number zero, rejection number 1.
Proof pressure	Critical	4.7.2	Every hose assembly	Acceptance number zero, rejection number 1.
Obstruction	Critical	4.7.3	Every hose assembly	Acceptance number zero, rejection number 1.
Endurance 2/ room temperature	Critical	4.7.4.1	Every hose assembly	Acceptance number zero, rejection number 1.
Impulse 2/ 3/	Major	4.7.6 (only 8000 cycles)	Inspection Level L-4 1/	Acceptance number zero, rejection number 1.
Preparation for delivery	Minor	4.7.1.2	Inspection Level L-4 1/	Total acceptable quality level of 4.0 percent defective.

1/ The sample size shall be based only on the applicable sample size code letter corresponding to the specified inspection level of Standard MIL-STD-105.

2/ Samples subjected to these destructive tests shall not be considered or shipped as part of the contract or order

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3/ For quality conformance inspection the hose assembly shall be subjected to 8000 cycles during the impulse test.

4.5.2.2 Sampling for tests and examinations of the hose assemblies at a laboratory - Upon completion of the tests and examinations specified in 4.5.2.1, a random sample shall be selected from each lot in accordance with Standard MIL-STD-105, Inspection Level L-6. The sample size shall be based only on the applicable size code letter corresponding to the Inspection Level L-6, Each hose assembly selected as a sample unit shall be forwarded to the Air Crew Equipment Laboratory or to a laboratory selected by the Air Crew Equipment Laboratory (see 6.4) for the following tests and examinations.

TESTS AND EXAMINATIONS

Visual examination	(4,7.1)
Dimensional check	(4.7.1.1)
Proof pressure	(4.7.2)
Obstruction	(4.7.3)
Impulse (only 8000 cycles)	(4.7.6)

The tests and examinations shall be conducted in the order listed. The acceptance number shall be zero, rejection number 1, for each test arid examination, except that the acceptable quality level for minor defects shall not exceed 10.0 defects per hundred units. The lot. from which the hose assembly samples were selected. shall not be shipped until the samples have been approved by the Air Crew Equipment Laboratory. The lot, from which the hose assembly samples were selected shall be rejected if any sample fails to comply with any requirement of this paragraph. A rejected lot shall not be resubmitted except with the approval of the contracting officer (see 6.4). Upon completion of the tests and examinations, the laboratory shall return the hose assembly samples to the manufacturer.

$$\text{Defects per hundred units} = \frac{\text{Number of defects x 100}}{\text{Number of units}}$$

4.6 Test conditions -

4,6.1 Gas - Unless otherwise specified, the gas used in testing the hose assemblies shall be oxygen conforming to Specification MIL-O-21749, type I.

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4.6.2 Temperature and pressure - Unless otherwise specified, tests shall be conducted at an ambient temperature of 77 ±18°F and at an ambient barometric pressure of 28 to 32 inches of mercury. If tests are conducted at different ambient conditions, appropriate correction factors shall be applied to the flow and pressure readings. All specified test pressures are gage pressures measured relative to the ambient air pressures,

4.7 Inspection methods -

4.7.1 Visual examination -

4.7.1.1 Hose - Every hose shall be checked dimensionally and examined visually to determine conformance to this specification and applicable drawings. The classification of defects. Tables II and III, shall be used to classify the defects found.

TABLE II

CLASSIFICATION OF DEFECTS FOR VISUAL EXAMINATION OF THE HOSE ASSEMBLY

CRITICAL	MINOR
1. Material imperfections - foreign matter embedded	201. Marking - missing, insufficient, incorrect, illegible, or not permanent
2. Surface - unclean, rough, misaligned, or containing cracks, nicks, or other flaws	202. Color not as specified
3. Any opening not capped	
4. Any component missing, malformed, fractured, or otherwise damaged	
5. Any component loose or otherwise not securely retained	
6. Incorrect assemblying or improper positioning of components	
7. Any functioning part that works with difficulty	
8. Faulty workmanship or other irregularities	

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TABLE III

CLASSIFICATION OF DEFECTS FOR FINISHED DIMENSIONS

EXAMINE	DEFECT
Measure hose and hose assembly	Any measurement deviating from the dimensions and applicable tolerances as specified in the specification and applicable drawings shall be classified as a dimensional defect

4.7.1.2 Preparation for delivery - Each of the fully prepared shipping containers containing hose assemblies, selected as a sample unit from the lot shall be visually examined to determine that the packaging, packing, and marking conform to this specification. The classification of defects, Table IV, shall be used to classify the defects found.

TABLE IV

LIST OF DEFECTS FOR PREPARATION FOR DELIVERY

ITEM	DEFECTS
Exterior and interior markings	Missing, incorrect, incomplete, illegible; of improper size, location, sequence; or method of application; markings not the same on the interior and exterior containers.
Packaging and Packing materials	Any non-conforming component; any component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure of the unit package, intermediate package, case liners, container flaps, loose strappings, etc.; bulging or distortion of the containers.
Exterior and interior weight or content	Number per container is more or less than required; gross or net weight exceeds the requirements; more than one type in the same container.

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4.7.2 Proof pressure - The hose assembly shall be immersed in water and subjected to an internal gaseous pressure of 450 psi and maintained for 5 minutes. There shall not be any evidence of leakage or damage.

4.7.3 Obstruction - A steel ball, 0.267 inches in diameter, shall be dropped through the internal passage of the hose assembly. The ball shall pass through without friction due to any obstruction.

4.7.4 Endurance -

4.7.4.1 Room temperature - The hose assembly shall be coiled around a mandrel so that the bend radius (inside of bend) of the hose is 3 inches. The coil shall be rolled the entire length of the hose, The hose assembly shall then be reversed and returned to the straight position. This procedure shall constitute one endurance cycle and shall be conducted for 250 cycles. The hose assembly shall then be subjected to and pass the proof pressure test and obstruction test.

4.7.4.2 Low temperature - The hose assembly shall be conditioned at -297° F or colder for 15 minutes. After conditioning, while still at this temperature, the hose assembly shall be subjected to 250 endurance cycles. The hose assembly shall then be subjected to and pass the proof pressure test and obstruction test,

4.7.5 Burst pressure - The hose assembly shall be subjected to a hydrostatic pressure (water shall be employed as the pressure medium) sufficient to burst the assembly with a rate of pressure rise equal to 20,000 ±5,000 psi per minute. The type of failure and the pressure required for failure shall be recorded. The assembly shall not leak at any pressure below 4800 psi.

4.7.6 Impulse - The hose assembly shall be subjected to a pressure impulse of 0 to 200 psi operating and 450 psi peak pressure, applied at 35 cycles per minute for 25,000 cycles (8000 cycles for quality conformance inspection). Water shall be employed as the pressure medium. The hose assembly shall then be subjected to and pass the proof pressure test.

4.7.7 Disassembly and examination - At the conclusion of all the inspections, the hose assembly shall be disassembled and examined. The hose assembly shall not show evidence of excessive deterioration or wear due to testing which might affect the hose assembly performance. The hose assembly shall be disassembled and examined after any test in which hose assembly failure indicates faulty construction.

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5. PREPARATION FOR DELIVERY

5.1 The hose assembly shall be prepared for delivery in accordance with Specification MIL-P-7936. Preservation for Level A shall be in accordance with Specification MIL-P-116, Method 1A, utilizing transparent film bags conforming to Specification MIL-B-22205, type II, without the use of contact preservatives and without cleaning by immersion in solvent. Sharp edges shall be cushioned and all openings shall be plugged with transparent film conforming to Specification MIL-F-22191. type I or type III,

5.1.1 Precautionary marking - The following precautionary marking shall appear on each unit package:

"CAUTION: DO NOT ALLOW CONTAMINANTS OF ANY KIND TO BE USED ON OR ABOUT THE OXYGEN TRANSFER HOSE."

5.1.2 The hose shall have a detachable marking tag wired to the hose advising users of maintenance and stationary safety precautions. The following items shall be included:

(1) Cleaned for oxygen service.

(2) Do not TWIST metal hose.

6, NOTES

6.1 Intended use - The metal, liquid oxygen hose assembly covered by this specification is intended for use in transferring liquid oxygen from a supply trailer to aircraft converter,

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number, and date of this document.

(b) Whether preproduction inspection is required (see 4.4.1).

(c) Selection of applicable levels of preservation, packaging, arid packing required.

(d) Applicable methods of cleaning and preservation,

(e) Name of laboratory for quality conformance inspection (see 4.5.2.2 and 6.4).

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6.3 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Chief, Bureau of Naval Weapons, Department of the Navy, Washington 25, D, C.; however. information pertaining to qualification of products may be obtained from the Director, Air Crew Equipment Laboratory, Naval Air Engineering Center, Philadelphia 12, Pennsylvania.

6.4 Laboratory information - The successful bidder will be furnished with the name of the inspection laboratory, if different from the Air Crew Equipment Laboratory (see 4.5 .2.2), at the time of the award. The cost of the tests and examinations on samples initially submitted from a lot, as required by 4.5.2.2, shall be borne by the Government whether the tests and examinations are conducted by the Air Crew Equipment Laboratory or by a laboratory selected by the Air Crew Equipment Laboratory. Samples from a rejected lot shall not be submitted for tests and examinations, as required by 4.5.2.2, without the approval of the contracting officer. The costs of the tests and examinations on samples submitted from a reworked lot or from a new lot which is necessitated by the rejection of a previous lot shall be borne by the manufacturer whether the tests and examinations are conducted by the Air Crew Equipment Laboratory or by a laboratory selected by the Air Crew Equipment Laboratory.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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:CIFICATION ANALYSIS SHEET
WEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 45-R309

INSTRUCTIONS

s sheet is to be filled out by Government or contractor sonnel involved in the use of this specification in curement of products for ultimate use by the activity wn on the reverse of this sheet.

s sheet is provided for obtaining information which l aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

IFICATION *(No. and abbrev. title)*

NIZATION	CITY	STATE

RACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

ERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT		OR A SUBCONTRACT		

IAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

OMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

S THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF YES, IN WHAT WAY?

MARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form* ic papers in an envelope. *This form is addressed for use in window envelope when appropriate.)*

ITTED BY *(Print name and title)*	DATE

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